Question					Marking details	Marks available					
						AO1	AO2	AO3	Total	Maths	Prac
11					**Indicative content** • H^+ and Na^+ ions attracted to negative electrode because opposites attract • H^+ ions gain electrons forming hydrogen (gas) • $2H^+ + 2e^- \rightarrow H_2$ • hydrogen formed rather than sodium because hydrogen is below sodium in reactivity series so Na^+ ions remain in solution • OH^- and Cl^- ions are attracted to the positive electrode because opposite attract • Cl^- ions lose electrons forming chlorine (gas) • $2Cl^- \rightarrow Cl_2 + 2e^-$ • OH^- ions less easily oxidised than Cl^- ions so remain in solution • Na^+ and OH^- ions remain in solution ⇨ sodium hydroxide	6			6		
					5-6 marks Full explanation of formation of hydrogen and chlorine with attempt at sodium hydroxide; good attempt at ionic equations *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** Good attempt at explanation of formation of hydrogen and chlorine; attempt at ionic equation *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Attempt at explanation of formation of hydrogen or chlorine *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*. *The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
					Question 11 total	**6**	**0**	**0**	**6**	**0**	**0**